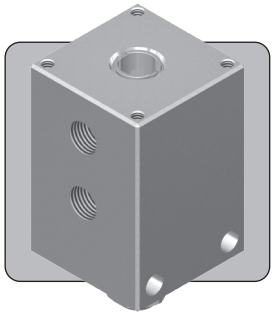
- 2/3DS25 -

SOLENOID VALVE

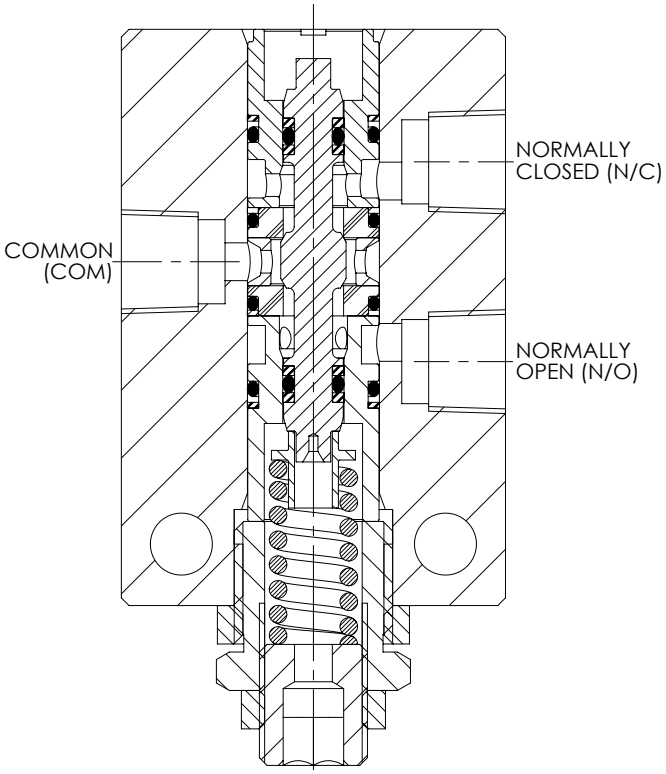
1/4" UP TO 414 BAR 6,000 PSI



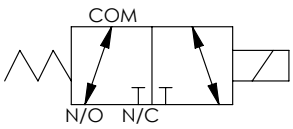
The 2/3DS25 is a 1/4" nominal bore 2 position 3 way direct acting solenoid valve. The valve uses a balanced spool and seat design, such that it may be used as either a normally closed or normally open valve and with bi-directional flow. The valve may also be used as a 2 port valve by plugging the unused port.

The valve uses soft seat technology, to give low leakage with liquid and gases.

- Stainless Steel Construction
- Soft seated design for low leakage
- Balanced spool design for bi-directional flow
- Approved hazardous area coil (EEx'd')
- Various thruster options
- Suitable for liquid and gas
- Suitable for use with air, nitrogen, sweet natural gas, mineral oils, water glycols, water and sea water
- Suitable for many other media, contact us for advice
- Various seal options available
- Various porting options available



Specifications

BASIC MODEL NUMBER	2/3DS25
SYMBOL	
MAX WORKING PRESSURE LIQUID	414 bar (6,000 psi)
MAX WORKING PRESSURE GAS	241 bar (3,500 psi)
CV (FLOW CAPACITY)	0.26
FLUID	Liquids and Gases See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/4"
WEIGHT	HC 2.6 kg (5.7 lb) KC 2.6 kg (5.7 lb) STEX1 4.0 kg (8.8 lb)

Specifications may change without notice

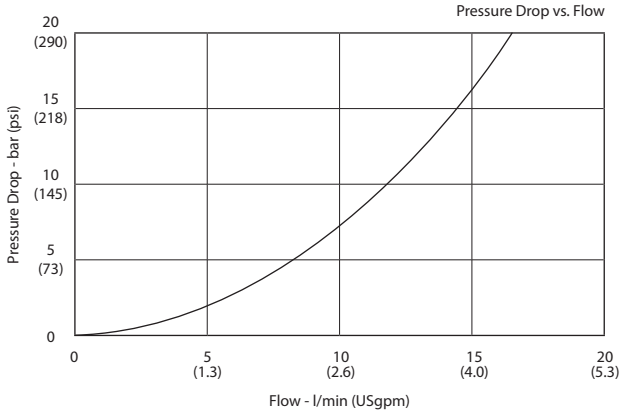
Materials

Externally Exposed Parts: 316 stainless steel with a 302 stainless steel spring. Refer to the relevant actuator data sheet for solenoid thruster materials.

Internally Wetted Parts: 316 stainless steel, PEEK, acetal and PTFE.
The standard valve is designed for use with air, nitrogen, sweet natural gas, mineral oils, water glycols, plain water and sea water and may also be used with a wide variety of media compatible with the materials of construction.

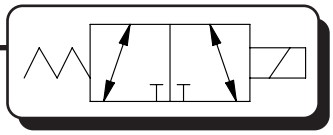
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

Typical Performance



Typical performance based on water

Solenoid Thruster Options



The 2/3DS25 is available with a number of solenoid thruster options, with various electrical connections and with certification for use in potentially explosive hazardous atmospheres.

HC Thrusters

The HC thruster is fitted with a DIN 43650 male plug base and is supplied complete with the female plug ready for connection, offering ingress protection to IP65. The HC thruster is not suitable for use in potentially explosive atmospheres.

KC Thrusters

The KC thruster is fitted with a 600mm (24 inch) long PVC covered steel pliable protective conduit terminated with a male 1/2" x 18 BS Conduit thread. Contact us for advice if a different thread size is required. The KC thruster is not suitable for use in potentially explosive atmospheres.

STEX1 Thrusters

The STEX1 series of solenoid thrusters are certified under the ATEX Directive (94/9/EC) for use in potentially explosive atmospheres. Using the Flameproof Ex 'd' protection concept, the thrusters are CE marked with the EX coding II2G, denoting them as category 2 products suitable for potentially explosive gas atmospheres in non-mining applications. As category 2 products, they may be used in zones 1 and 2, as defined and assessed under EC Directive 1999/92/EC. Depending on the ambient temperature (see table below), the thrusters are certified for use where T4 or T5 equipment surface temperatures are allowed, including in T1, T2 and T3 environments. The electrical wiring used to power the product should have an environmental temperature rating at least 55°C above the maximum ambient temperature, and suitable protective conduit should be installed in the M20 x 1.5 threaded connection port.

THRUSTER TYPE	POWER CONSUMPTION	AMBIENT TEMPERATURE RANGE	ATEX TEMPERATURE CLASS
HC	15.6 Watts	-10˚C to +60˚C	Not applicable
KC	15.6 Watts	-10˚C to +60˚C	Not applicable
STEX1	15.8 Watts	-20˚C to +40˚C	T5 (100˚C)
		-20˚C to +60˚C	T4 (135˚C)

Note: Interflow exists during switching

Product Selector

2/3DS25 P - G - HC - NBR - 6K - 24VDC

VALVE TYPE	PORTING OPTIONS	SEAT TYPE	THRUSTER OPTIONS	SEAL OPTIONS	MAX WORKING PRESSURE	COIL VOLTAGE
		G option must be selected		Leave blank for Viton® seals -10˚C to +80˚C		
2/3DS25	P 1/4" BSPP female	G Soft seats	HC Din connector	NBR Nitrile seals -30˚C to +80˚C	6K 414 bar (6,000 psi) Liquid	24VDC 24 Volt DC
	N 1/4" NPT female		KC Kopex fitting	KLZ Kalrez® seals -25˚C to +80˚C	241 bar (3,500 psi) Gas	Further voltage options are available on request
	M Manifold mount		STEX1A EEx'd' approved Side cable entry	HNBR Hydrogenated Nitrile seals -20˚C to +80˚C		
			STEX1B EEx'd' approved Top cable entry	EP EPDM seals -45˚C to +80˚C		
				Further seal options are available on request		
				Note that the seal temperature ranges relate to fluid temperature. For allowable ambient temperatures refer to the thruster table above		

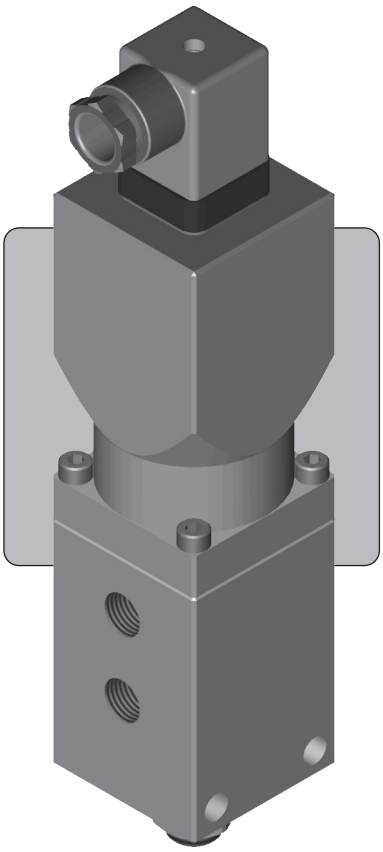
Repair and seal kits are available on request

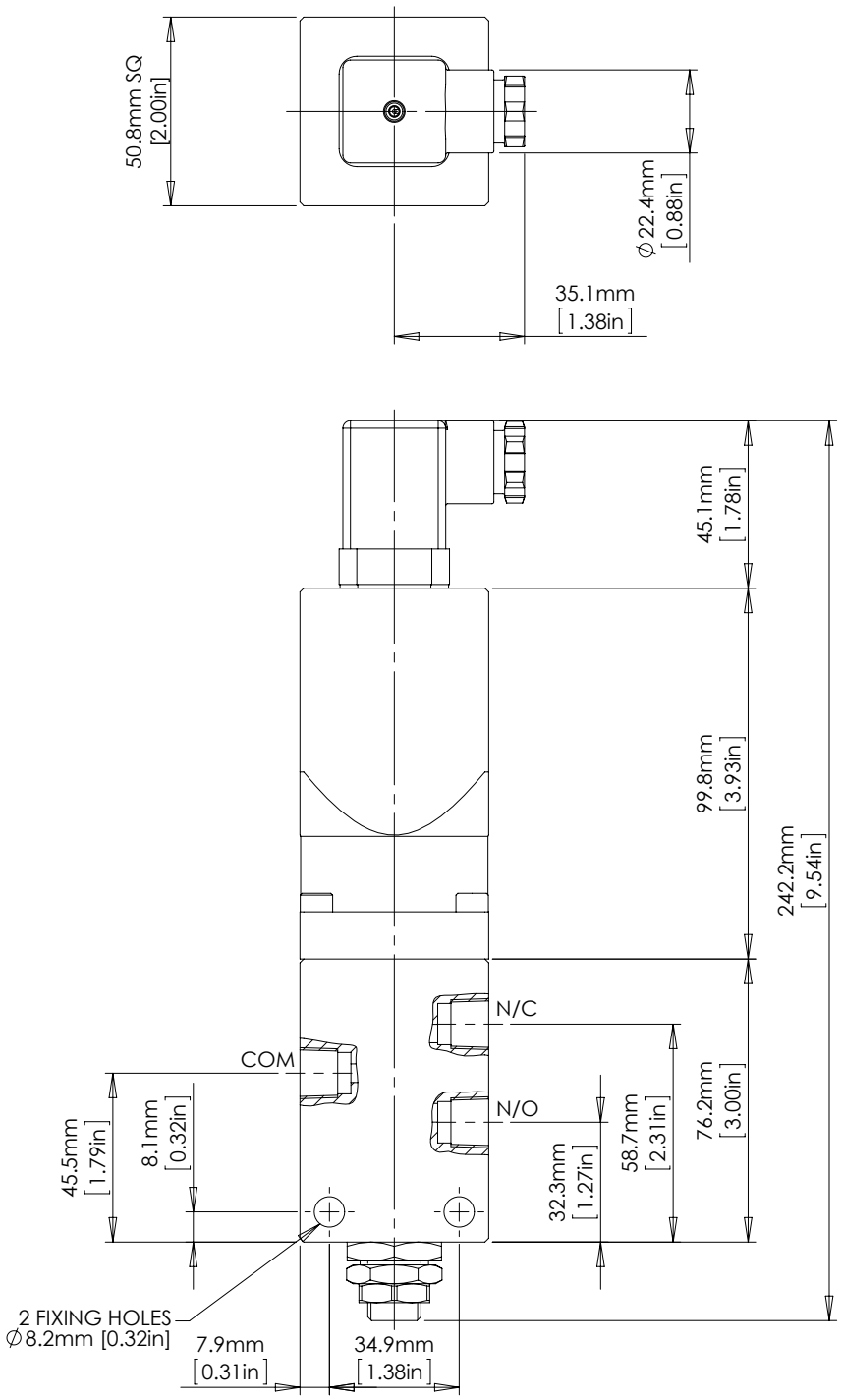
For a seal kit add SK at the end of the model code and for a repair kit add RK

2/3DS25

SOLENOID VALVE

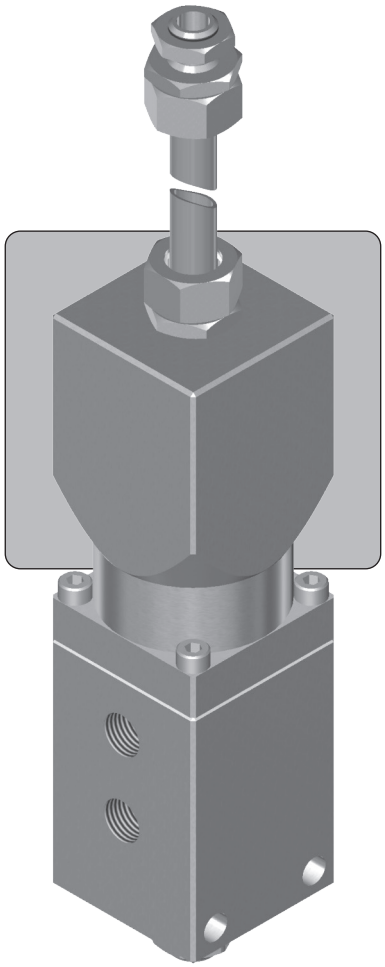
2/3DS25-HC

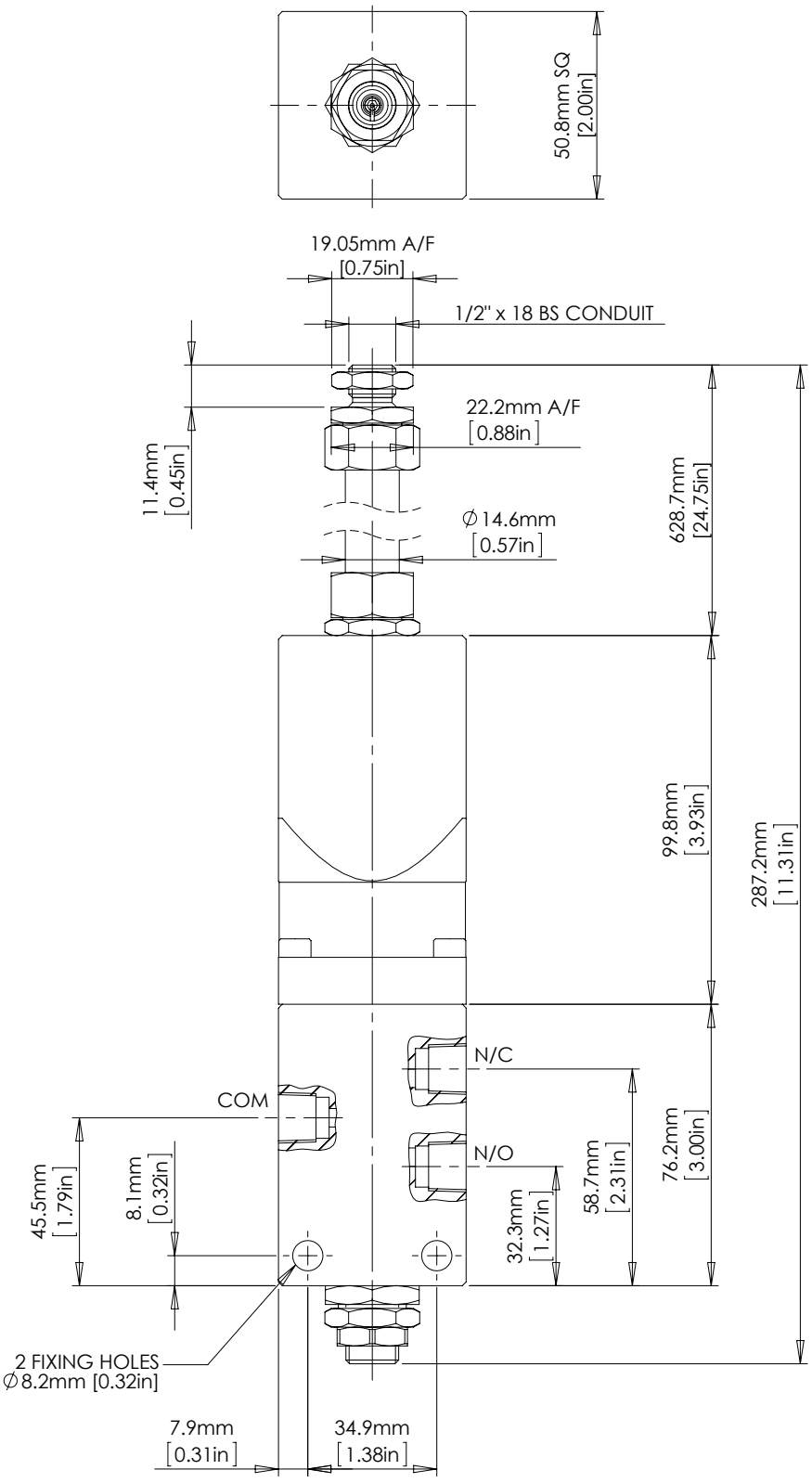




BS VALVES

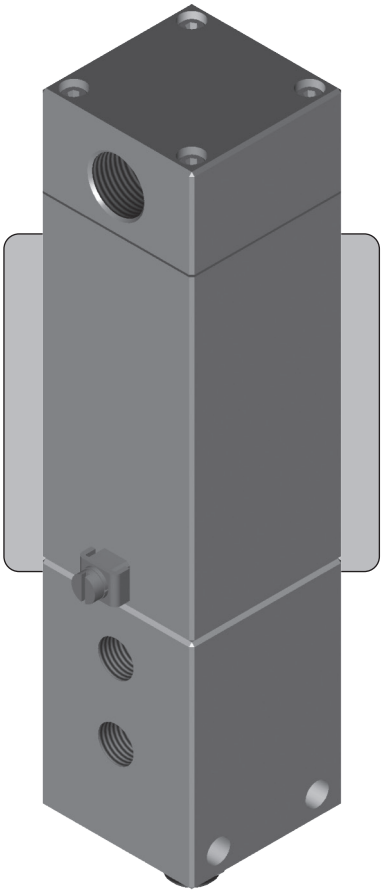
2/3DS25-KC

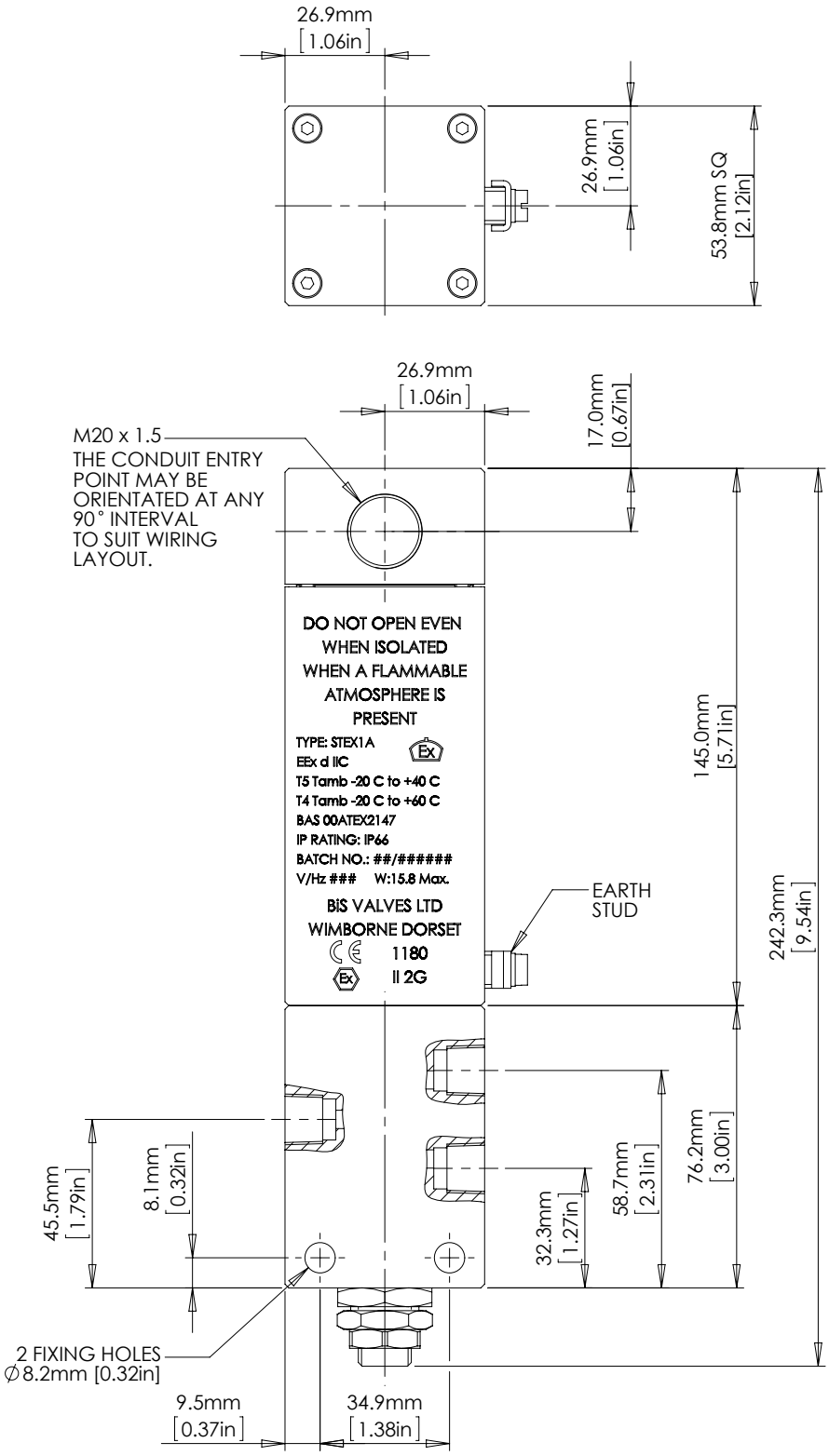



50.8mm SQ
[2.00in]
19.05mm A/F
[0.75in]
1/2" x 18 BS CONDUIT
22.2mm A/F
[0.88in]
11.4mm
[0.45in]
∅14.6mm
[0.57in]
628.7mm
[24.75in]
99.8mm
[3.93in]
287.2mm
[11.31in]
N/C
COM
N/O
45.5mm
[1.79in]
8.1mm
[0.32in]
32.3mm
[1.27in]
58.7mm
[2.31in]
76.2mm
[3.00in]
2 FIXING HOLES
∅8.2mm [0.32in]
7.9mm
[0.31in]
34.9mm
[1.38in]

2/3DS25
SOLENOID VALVE

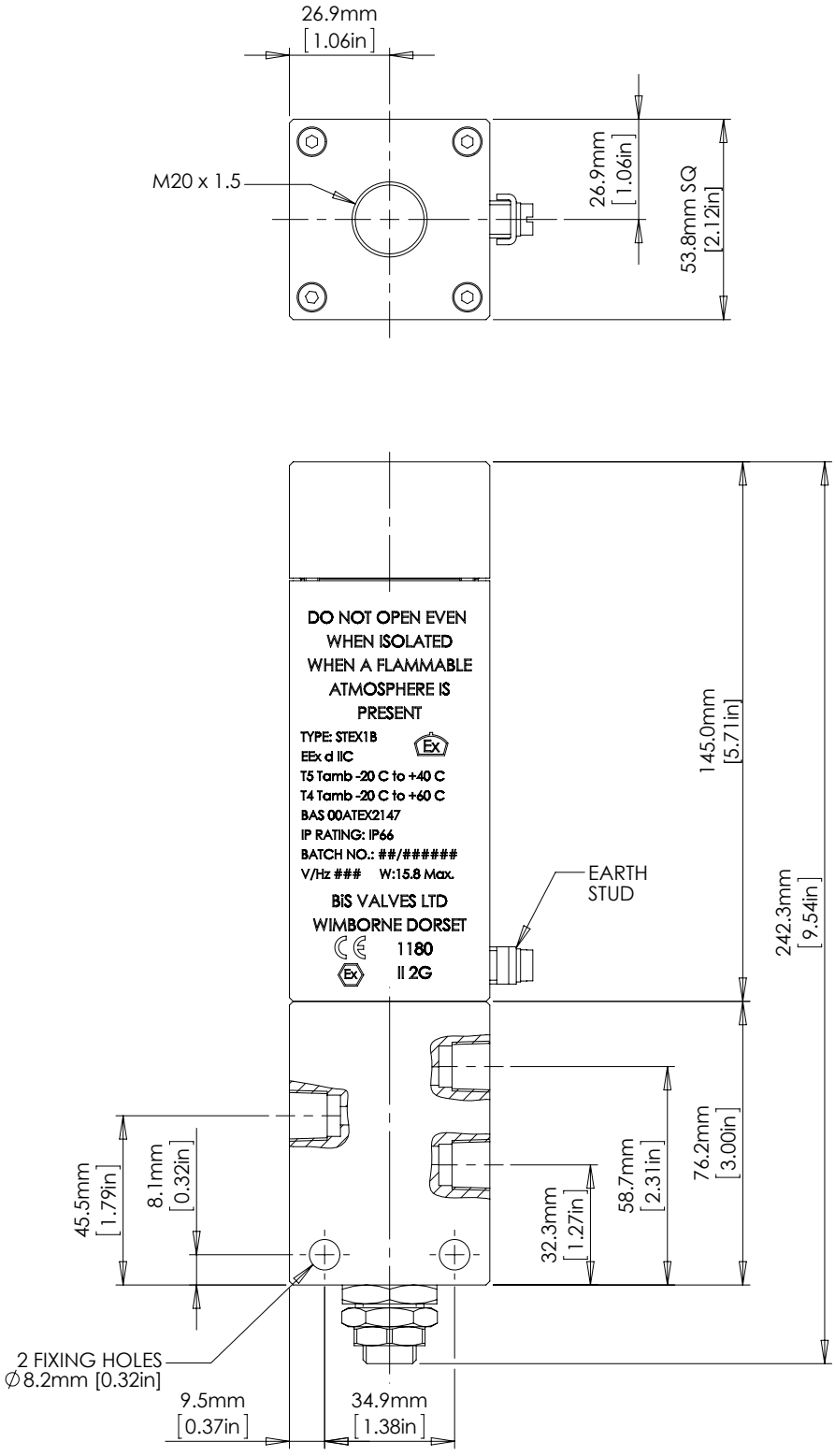
2/3DS25-STEX1A





2/3DS25-STEX1B

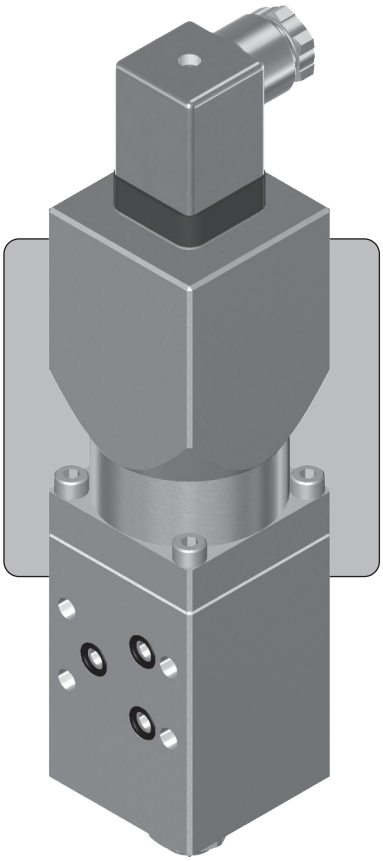


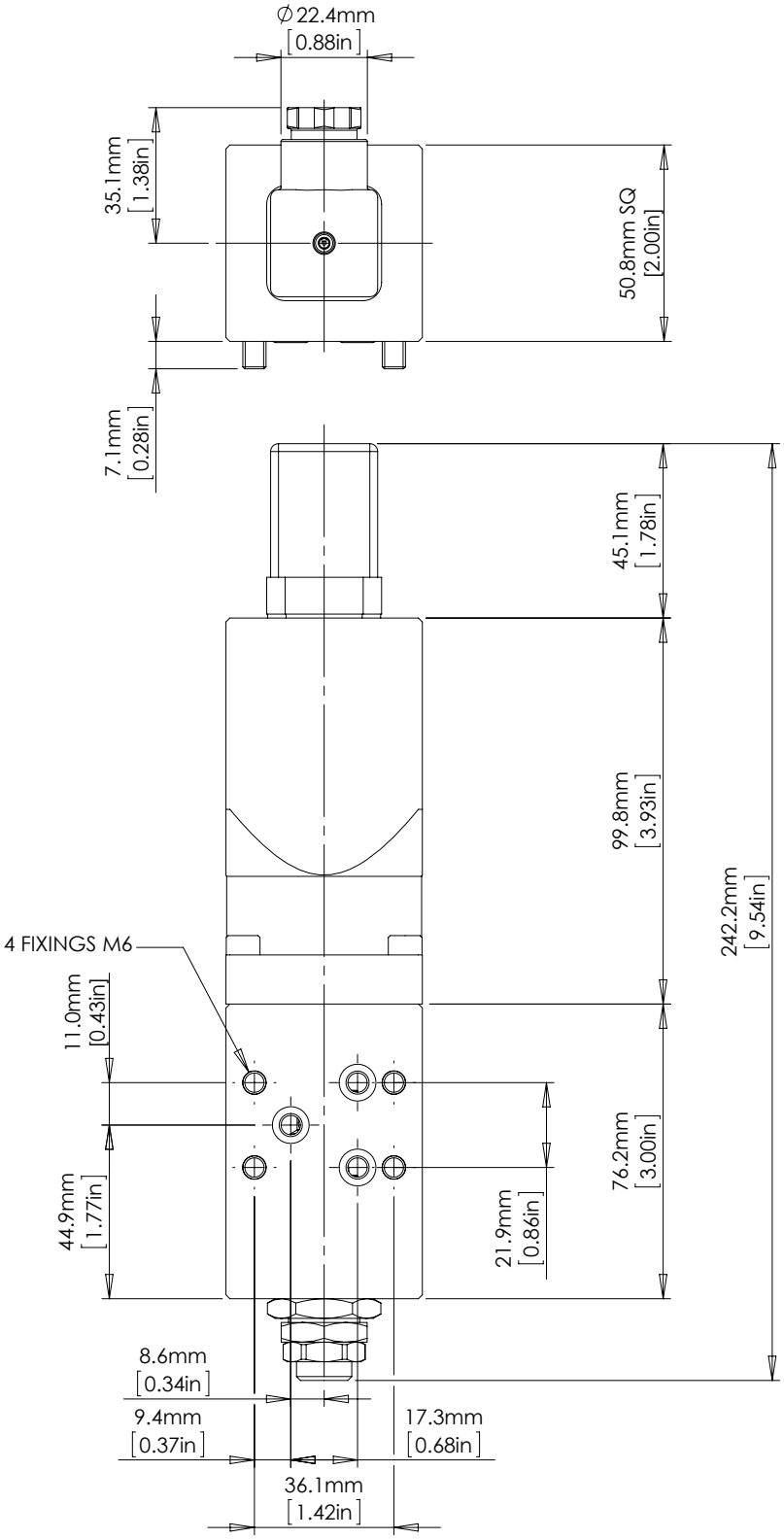


2/3DS25

SOLENOID VALVE

2/3DS25M-HC





2/3DS25M-STEX1A



